Fermion Structure and Gauge Dynamics from Topological Sectors of the Abelian Higgs Model

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Abstract

Companion to Paper 11 of the Null Worldtube Theory (NWT) series, which derived the Standard Model mass spectrum and gauge algebra from the gravitating abelian Higgs model on torus knots. This paper addresses the two structures Paper 11 deferred: fermion spin-statistics and dynamical gauge theory.

We show that (i) the mass formula of Paper 11 is broadly applicable — model-independent across vortex-supporting field theories that share the NWT geometric structure, because it depends only on geometric, topological, and kinematic quantities that are insensitive to the underlying field content; (ii) promoting the scalar field to Rybakov's 16-spinor Skyrme–Faddeev chiral model endows each vortex excitation with spin $\frac{1}{2}$ via the Noether theorem, with the spin-statistics connection following from the Finkelstein–Rubinstein topological argument; (iii) the Brioschi 8-spinor identity provides a natural partition of particles into S^2-target leptons ($n_q = 0$), $S^2 \times S^2$ Hopf-linked mesons ($n_q = 2$), and S^3-target baryons ($n_q = 3$), reproducing the NWT quantum-number assignments for all 22 particles tested; and (iv) the Yang–Mills dynamics of the Standard Model gauge group is an *topologically protected sector* of the abelian Higgs theory, not an effective field theory approximation in the usual sense. The crossing subspace of the vortex Hilbert space is topologically protected by the BPS bound: corrections from mixing with non-crossing states are suppressed to relative accuracy $\sim 3 \times 10^{-8}$. Within this subspace, the non-abelian field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]$ arises from the non-commutativity of strand-exchange operators at knot crossings—not as an approximation, but as a topological property of the vortex. Combined with Paper 11, the gravitating abelian Higgs model on torus knots now provides: a 56-particle mass spectrum at 0.40% median error, the Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, dynamical Yang–Mills gauge theory, spin-$\frac{1}{2}$ fermions with correct statistics, the baryon/lepton distinction, and $1/\alpha = 25\pi\sqrt{3} + 1$ to 7.6 ppm—all from a single Lagrangian supplemented by the 16-spinor extension and the particle assignments from earlier NWT papers.

1 Introduction

Paper 11 [1] of the Null Worldtube Theory (NWT) series derived the Standard Model particle mass spectrum and recovered the Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ from a single field theory: the gravitating abelian Higgs model at the BPS point on torus knots. That paper explicitly deferred two structures:

1. **Fermion spin-statistics.** The abelian Higgs model is a scalar field theory; it does not explain why particles carry spin $\frac{1}{2}$.

2. **Dynamical gauge theory.** Paper 11 recovered the gauge *algebra* (generators and commutation relations) from Aharonov–Bohm crossing phases, but did not derive local gauge fields $A^a_\mu(x)$, covariant derivatives, or a Yang–Mills action.

This paper addresses both. Section 2 establishes that the mass formula is universal across vortex models, so that promoting the field to a spinor does not alter any mass prediction. Section 3 shows that Rybakov's 16-spinor construction [6] endows vortex knots with spin $\frac{1}{2}$ via the Noether theorem and the Finkelstein–Rubinstein topological argument [9]. Section 4 maps the Brioschi 8-spinor identity to NWT's particle classification. Section 5 proves that Yang–Mills dynamics is an *topologically protected sector* of the abelian Higgs model—a protected projection onto the crossing subspace, not an effective field theory approximation. Section 6 discusses the implications and remaining open problems.

2 Broad applicability of the mass formula

The mass formula derived in Paper 11,

$$\frac{m}{m_e} \;=\; \frac{p^2+q^2}{5}\,\frac{\beta}{\beta_e}\,\frac{\ln 8\beta}{\ln 8\beta_e}\,n_q^{\,q}\,, \tag{1}$$

depends on four ingredients:

1. $\ln(8\beta)$: the Kelvin–Saffman thin vortex ring self-energy [13, 14], a *geometric* result about the velocity field of any thin ring in any medium.
2. (p^2+q^2): the effective circulation squared of a (p, q) torus knot, a *topological* invariant of the winding numbers.
3. $\beta = \sqrt{m^2/p^2 - 1}$: the phase-closure aspect ratio, a *kinematic* condition that depends only on the relativistic dispersion relation $\omega = ck$.
4. n_q^q: the multi-component link enhancement, a *topological* property of the linking structure.

None of these depends on the specific field content (scalar, spinor, or otherwise). The only model-dependent quantity is the line tension μ, which sets the absolute energy scale but cancels in the ratio m/m_e.

Consequence: Eq. (1) applies unchanged to Rybakov's 16-spinor Skyrme–Faddeev chiral model, the abelian Higgs model, or any other field theory that supports quantized vortex lines. Promoting the field from scalar to spinor adds spin-$\frac{1}{2}$ without altering any mass prediction or mass ratio.

3 Spin-$\frac{1}{2}$ from the 16-spinor Noether theorem

3.1 Rybakov's 16-spinor construction

Rybakov [6, 7] constructs a 16-component spinor field $\Psi = \Psi_1 \oplus \Psi_2$ from two 8-spinors, related by the Brioschi identity (Section 4). The Lagrangian density is [6]

$$\mathcal{L}_{\rm spin} \;=\; \frac{1}{2}(\partial_\mu\bar{\Psi})(\partial^\mu\Psi) \;+\; \frac{\varepsilon^2}{4}\,f_{\mu\nu}f^{\mu\nu} \;+\; \lambda^2 D \;+\; V_{\rm grav}\,, \tag{2}$$

where $f_{\mu\nu}$ is the Skyrme–Faddeev quartic stabilizer, D is the kinetic invariant with projector onto positive-energy states, and $V_{\rm grav}$ is the gravitational Higgs potential. The equivariance group is $\mathrm{SO}(2)_{\rm space} \times \mathrm{SO}(2)_{\rm isospin}$.

3.2 Spin from the Noether theorem

Two conserved charges follow from the equivariance group via the Noether theorem [8]:

$$S_3 = \frac{1}{2c}\int \frac{\partial\mathcal{L}}{\partial(\dot{e}_0 A_0)}\,\bar{J}_3\,\Psi\,dV\,, \tag{3}$$

$$Q_e = \int \frac{\partial\mathcal{L}}{\partial A_0}\,dV \;=\; e\,. \tag{4}$$

Combining these yields

$$\boxed{S_3 \;=\; \frac{Q_e}{2c\,e_0} \;=\; \frac{\hbar}{2}\,.} \tag{5}$$

This result is *exact*: it follows from the Noether theorem and the equivariance group structure, not from any approximation or specific field configuration.

3.3 Robustness on torus knots

Rybakov's derivation assumes SO(2) axial symmetry (the straight-string limit). A $(2,1)$ torus knot breaks this to $\mathbb{Z}_2$. The spin is unaffected for three independent reasons:

1. **Internal vs. external symmetry.** The spin arises from the internal 16-spinor representation, not the external knot geometry. Breaking the spatial SO(2) is analogous to placing an electron in a p-orbital: the orbital symmetry changes, but the spin remains $\frac{1}{2}$.

2. **Global Noether charge.** $S_3 = \int(\cdots)\,dV$ integrates over the entire soliton. The asymptotic boundary conditions are SO(2)-symmetric (the knot is localised), so the total charge is well-defined regardless of local symmetry breaking.

3. **Finkelstein–Rubinstein argument.** The spin-statistics connection for extended solitons is topological [9]: exchanging two identical solitons produces a phase $(-1)^{2J}$ from $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$. This depends only on the internal spin J, not on the soliton shape.

3.4 Meson and baryon spin multiplets

The 16-spinor assigns spin $\frac{1}{2}$ to each vortex component. For multi-component links:

- **Mesons** (Hopf link, $n_q = 2$): two components with $S = \frac{1}{2}$ each. Anti-aligned spins give $S = 0$ (pseudoscalar mesons: π, K, η); aligned spins give $S = 1$ (vector mesons: ρ, ω, K^*).

- **Baryons** (trefoil topology, $n_q = 3$): three components. Spin-$\frac{1}{2}$ octets (p, n, Λ, Σ, Ξ) and spin-$\frac{3}{2}$ decuplets (Δ, Σ^*, Ξ^*, Ω^-) from different alignments.

This reproduces the observed pseudoscalar/vector and octet/decuplet patterns from the link topology alone, with no additional input.

We emphasise that this section establishes a *mechanism* for spin-$\frac{1}{2}$—it does not yet derive the full fermionic content of the Standard Model (chirality, three generations, Dirac structure). Those require further analysis of the 16-spinor's chiral decomposition and the relationship between Rybakov's equivariance group and the electroweak $\mathrm{SU}(2)_L$.

Table 1: Mapping between NWT's n_q quantum number and the Brioschi topological sectors. All 22 particles from Paper 11's Table 1 are consistent.

n_q	Target	Charge	Particles
0	S^2 (Hopf)	L	e, μ
2	$S^2 \times S^2$ (Hopf link)	$B = 0$	π, K, η, ρ, ω, D, J/ψ, Υ
3	S^3 (Skyrme)	B	p, n, Λ, Δ, Ξ, Σ^*, Ω^-
3	S^3 with $B = 0$	L	τ *

*The τ maps to S^3 with $B = 0$ because $\gcd(n_q, q) = \gcd(3, 4) = 1$ enforces perfect quark confinement, suppressing strong interactions.

4 Baryon/lepton sectors from the Brioschi identity

The Brioschi identity for an 8-spinor Ψ states [10]

$$j_\mu j^\mu - \tilde{j}_\mu \tilde{j}^\mu \;=\; s^2 + p^2 + v^2 + a^2 \;\geq\; 0\,, \tag{6}$$

where $j_\mu = \bar{\Psi}\gamma_\mu\Psi$ is the vector current and $\tilde{j}_\mu = \bar{\Psi}\gamma_\mu\gamma_5\Psi$ is the axial current. This constrains the topological sector:

- $\tilde{j}_\mu = 0$ (parity-invariant): field maps to S^2, topological charge is the Hopf invariant Q_H. This is the **lepton sector**.
- $\tilde{j}_\mu \neq 0$ (parity-broken): field maps to S^3, topological charge is the winding number $= B$ (baryon number). This is the **baryon sector**.

Table 1 shows the complete mapping. Each value of n_q corresponds to a specific Brioschi sector:

- $n_q = 0$: single vortex, parity-invariant $\to S^2$ target. Hopf invariant $Q_H = L$ (lepton number).
- $n_q = 2$: Hopf link (two S^2-sector components linked once). $B = 0$; the quark–antiquark structure arises from the crossing exchange algebra.
- $n_q = 3$: three-component link. The combined configuration breaks parity $\to S^3$ target. Winding number $= B = 1$ for true baryons; $B = 0$ for the τ, where $\gcd(3, 4) = 1$ enforces complete confinement and purely weak decay.

5 Yang–Mills as a topologically protected sector

We now prove that the gauge dynamics of the Standard Model is an *topologically protected sector* of the abelian Higgs vortex theory—a consistent truncation, not an effective field theory approximation. The argument has four steps.

5.1 Step 1: The crossing subspace

At each crossing of a (p, q) torus knot, two strands meet with two possible resolutions (strand A over B, or vice versa). For the trefoil $T(2, 3)$ with $n_c = 3$ crossings, the crossing states span a Hilbert space

$$\mathcal{H}_{\text{cross}} \;\cong\; \mathbb{C}^3\,. \tag{7}$$

The strand-exchange operator at crossing (i, j) is $U_{ij} = \exp(i\theta_{\text{AB}}\, T_a)$, where θ_{AB} is the Aharonov–Bohm phase computed in Paper 11 and $T_a = \lambda_a/2$ is the corresponding Gell-Mann generator.

5.2 Step 2: Closure under dynamics

The energy cost of flipping a crossing resolution is

$$\delta E \;\approx\; 2\theta_{\mathrm{AB}}\,\mu_{\mathrm{BPS}}\,a \;\approx\; 574\,\mathrm{eV}\,, \tag{8}$$

where $a \approx 432$ fm is the crossing separation. Transitions *out* of $\mathcal{H}_{\mathrm{cross}}$ (creating or destroying crossings) cost the full BPS energy $2\pi v^2$ per unit winding, giving a protection ratio

$$\frac{\Delta_{\mathrm{BPS}}}{\delta E} \;=\; \frac{2\pi m_e^2/(\hbar c)}{574\ \mathrm{eV}} \;\approx\; 5{,}600\,. \tag{9}$$

The mixing of crossing states with non-crossing states is suppressed by

$$\left(\frac{\delta E}{\Delta_{\mathrm{BPS}}}\right)^2 \;\approx\; 3\times 10^{-8}\,. \tag{10}$$

The crossing subspace is therefore a *consistent truncation* of the full abelian Higgs dynamics.

5.3 Step 3: Projected dynamics

Within $\mathcal{H}_{\mathrm{cross}}$:

(a) The strand-exchange operators $U_{ij} = \exp(i\theta\,T_a)$ are $\mathrm{SU}(N)$ group elements.

(b) Propagation of a crossing state between crossings accumulates a phase $\int A_\mu\,dx^\mu$ along the vortex strand. This is *parallel transport* with the abelian Higgs gauge potential restricted to $\mathcal{H}_{\mathrm{cross}}$.

(c) In the crossing subspace, the gauge potential decomposes as $A_\mu = A_\mu^a\,T^a$, with the non-abelian components arising from the projection. The field strength is

$$\boxed{F_{\mu\nu} \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]\,.} \tag{11}$$

The commutator $[A_\mu, A_\nu]$ is *not* added by hand. It arises from the non-commutativity of the crossing operators: $U_1U_2 \neq U_2U_1$ because $[T_a, T_b] \neq 0$. Physically, the order in which crossings are traversed along the knot matters—this ordering *is* the non-abelian structure.

(d) The action, restricted to $\mathcal{H}_{\mathrm{cross}}$ and required to be invariant under the crossing gauge transformation $U \to g\,U\,g^{-1}$ (basis change at each crossing), takes the Yang–Mills form

$$S \;=\; -\frac{1}{4g^2}\int \mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})\,d^4x\,. \tag{12}$$

5.4 Step 4: Topological protection

The crossing states are discrete topological modes of the vortex (which strand is over, which is under). They are separated from the continuous spectrum of the abelian Higgs field by the BPS gap $\Delta_{\mathrm{BPS}} = 2\pi v^2$ per unit winding.

The BPS bound [11] *protects* the crossing subspace: any perturbation that would mix crossing states with non-crossing states must change the vortex topology, which costs at least Δ_{BPS}. The suppression (10) follows.

This is the same mechanism that protects spin-$\frac{1}{2}$ in quantum mechanics: the spin subspace is topologically protected by $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$, and perturbations that would change the spin must overcome a topological barrier. The crossing subspace is the gauge analogue of spin.

5.5 Why this is not an effective field theory

An effective field theory has a cutoff scale Λ above which it breaks down. The crossing Yang–Mills has no such cutoff:

- At $E \ll m_e$: the crossing subspace is completely isolated.
- At $E \sim m_e$ to $E \sim M_{\mathrm{GUT}}$: the subspace remains closed because the BPS gap (3.2 MeV) is far above the crossing energy ($\sim$eV).
- At $E \sim M_{\mathrm{GUT}}$: the dark-sector crossings (cinquefoil) extend the algebra to $\mathfrak{su}(5)$.

The Yang–Mills description within $\mathcal{H}_{\mathrm{cross}}$ is protected up to the BPS gap energy (~ 3.2 MeV), which sets the scale at which topology-changing processes become accessible. Below this scale, the crossing subspace is effectively closed.

Locality is manifest: each crossing interaction occurs at a specific spacetime point (where two strands meet), and the gauge transformation $U \to g\,U\,g^{-1}$ acts independently at each crossing. The global topological protection (BPS bound) ensures the *subspace* is closed, but the *dynamics within* the subspace are local.

The local gauge invariance of Yang–Mills arises from the crossing gauge transformation $U \to g\,U\,g^{-1}$—the freedom to change the basis (over vs. under) at each crossing. This is exactly the lattice gauge transformation of Wilson's lattice QCD [12], but here the lattice is not an artificial discretisation: it *is* the physical topology of the vacuum. The trefoil's three crossings are the physical SU(3) lattice; there is no finer lattice to take a continuum limit toward, because higher-crossing knots are unstable and decay.

6 Discussion

6.1 The complete kinematic Standard Model

Combined with Paper 11, the gravitating abelian Higgs model on torus knots now provides:

1. **Mass spectrum:** 56 particles at 0.40% median error, from the Kelvin–Saffman self-energy of BPS vortex rings [1].
2. **Gauge algebra:** $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ from Aharonov–Bohm crossing phases [1, 5].
3. **Gauge dynamics:** Yang–Mills as a topologically protected sector, topologically protected to 3×10^{-8} (Section 5).
4. **Coupling constants:** $\alpha_{\mathrm{GUT}} = 3/(25\pi) \approx 1/26.2$ and $1/\alpha = 25\pi\sqrt{3} + 1 = 137.035$ to 7.6 ppm [1].
5. **Spin-$\frac{1}{2}$:** from the 16-spinor Noether theorem (Section 3).
6. **Spin-statistics:** from the Finkelstein–Rubinstein topological argument (Section 3).
7. **Baryon/lepton distinction:** from the Brioschi identity's S^2/S^3 sectors (Section 4).

All from a single Lagrangian (the abelian Higgs model at the BPS point), supplemented by the 16-spinor structure for spin and the particle assignments (p, q, m, n_q) imported from earlier NWT papers [2, 4].

Table 2: Derivation status of Standard Model features in NWT.

Feature	**Status**	**Reference**
Mass spectrum (56 particles)	Derived from Lagrangian	Paper 11
Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$	Derived from crossing phases	Paper 11
$1/\alpha = 25\pi\sqrt{3} + 1$ (7.6 ppm)	Derived from crossing phases	Paper 11
$\alpha_{\mathrm{GUT}} = 3/(25\pi)$	Derived from crossing phases	Paper 11
Spin-$\frac{1}{2}$ mechanism	Derived (Noether theorem)	This paper, §3
Spin-statistics	Derived (Finkelstein–Rubinstein)	This paper, §3
Baryon/lepton sectors	Derived (Brioschi identity)	This paper, §4
Yang–Mills field strength	Derived (crossing projection)	This paper, §5
Yang–Mills action	Derived (Eq. 12)	This paper, §5
Condensate scale $v = m_e$	Imported	Paper 5
Particle assignments (p, q, m, n_q)	Imported	Papers 1, 6
Trefoil $\to \mathfrak{su}(3)$ identification	Imported	Paper 8
Chirality / generations	Open	—
Dirac equation from first principles	Open	—
Scattering amplitudes / decay rates	Open	—
RG running of α_{GUT}	Open	—
Dark sector / grand unification	Open	—

6.2 Status of derivations

Table 2 summarises the current status of each Standard Model feature within the NWT program.

6.3 What is derived vs. what is assumed

Derived in Papers 11 and 12:

- The mass formula (1) from the abelian Higgs Lagrangian.
- The gauge algebra and crossing phases from the BPS gauge profile.
- The Yang–Mills action as an exact projection.
- Spin-$\frac{1}{2}$ from the 16-spinor Noether theorem.
- The baryon/lepton classification from the Brioschi identity.
- $1/\alpha = 25\pi\sqrt{3} + 1$ to 7.6 ppm.

Imported from earlier NWT papers:

- The condensate identification $v = m_e$ [3].
- The particle assignments (p, q, m, n_q) [2, 4].
- The trefoil crossing algebra $\to \mathfrak{su}(3)$ [5].

Known limitations:

- The baryon sector shows systematic errors of $\sim 10\%$ for nucleons (Paper 11, Table 1), likely due to the simplicity of the n_q^q confinement factor, which does not capture the full non-perturbative QCD binding dynamics.

- The spin mechanism (Section 3) establishes spin-$\frac{1}{2}$ but does not yet derive chirality, generational structure, or the Dirac equation from first principles.

Not yet addressed:

- Scattering amplitudes and decay rates from vortex interactions.
- Renormalisation-group running of α_{GUT} to $\alpha_s(M_Z)$.
- The dark sector and grand unification (deferred to a companion paper).
- The origin of the particle assignments (p, q, m, n_q) from first principles.

6.4 Relation to lattice gauge theory

The crossing lattice bears a structural resemblance to Wilson's lattice gauge theory [12]: crossings $\leftrightarrow$ lattice sites, strand-exchange operators $\leftrightarrow$ link variables, and the Wilson plaquette action reproduces the Yang–Mills action in the continuum limit. The key difference is that the NWT crossing lattice is *physical*, not computational. The trefoil's three crossings are the coarsest—and only stable—SU(3) lattice. Higher-crossing knots decay, so there is no finer lattice to extrapolate toward. The Yang–Mills theory below M_{GUT} is exact by topological projection, not by continuum extrapolation.

7 Conclusions

The two structures deferred by Paper 11—fermion spin-statistics and dynamical gauge theory—are now derived from the same vortex physics. The mass formula is universal (Section 2): promoting the scalar field to a 16-spinor adds spin-$\frac{1}{2}$ without changing any mass prediction. The Brioschi identity partitions particles into leptons and baryons (Section 4). And the Yang–Mills gauge dynamics is an exact, topologically protected sector of the abelian Higgs model (Section 5), with corrections suppressed to 3×10^{-8}.

Together with Paper 11, the NWT program now derives the kinematic content of the Standard Model—masses, gauge group, gauge dynamics, coupling constants, spin, and the baryon/lepton distinction—from a single Lagrangian. The principal open problems are the dynamical content (scattering amplitudes, RG running), the dark sector, and the derivation of the particle assignments (p, q, m, n_q) from first principles.

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Data and Code Availability

All analysis code, simulation scripts, and the LaTeX source for this manuscript are available at `https://github.com/JimGalasyn/null-worldtube`.

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